

1 Surface Tension of 316L with PLC Contamination

50 μm PLC ($\text{C}_6\text{H}_{10}\text{O}_2$; $f_C = 63.2 \text{ wt}\%$, $f_O = 28.1 \text{ wt}\%$, hydrogen lost as gas) coating dissolved into a 500 μm melt pool at $T = 1873 \text{ K}$ [1].

1.1 Notation

- f_C, f_O — elemental mass fractions of carbon and oxygen *in the PLC polymer itself* (i.e. the chemical composition of the coating before it contacts the melt). For $\text{C}_6\text{H}_{10}\text{O}_2$ with hydrogen lost as gas: $f_C = 72/114 = 63.2 \text{ wt}\%$, $f_O = 32/114 = 28.1 \text{ wt}\%$.
- w_C, w_O — weight-percent concentrations of C and O *dissolved in the steel melt* after the PLC has mixed in. These are the quantities that enter the surface tension model.
- w_C^0, w_O^0 — baseline C and O in virgin 316L stainless steel before any PLC contact ($w_C^0 \approx 0.030 \text{ wt}\%$, $w_O^0 \approx 0.001 \text{ wt}\%$).
- x_{PLC} — effective mass fraction of PLC dissolved into the melt, accounting for dissolution efficiency η .
- $\eta \in [0, 1]$ — dissolution efficiency: fraction of PLC that actually mixes into the melt; the remainder pyrolyses to gas or is lost as CO outgassing. A value $\eta = 0.1$ is adopted.

1.2 Dissolution Efficiency

$$w_{C,O} = x_{\text{PLC}}(\eta) f_{C,O} + (1 - x_{\text{PLC}}) w_{C,O}^0, \quad x_{\text{PLC}} = \frac{\eta \rho_{\text{PLC}} h}{\eta \rho_{\text{PLC}} h + \rho_s D}, \quad (1)$$

with $\rho_{\text{PLC}} = 1.145 \text{ g cm}^{-3}$, $\rho_s = 8.0 \text{ g cm}^{-3}$, $h = 50 \mu\text{m}$, $D = 500 \mu\text{m}$.

1.3 Surface Tension Model

$$\frac{\sigma}{\text{mN m}^{-1}} = \underbrace{1925 - 0.455 (T - 1808)}_{\text{pure Fe}} - 0.155 T \ln \left[1 + \underbrace{10^{-0.17 w_O - 0.427 w_C} w_O \exp\left(\frac{29300}{T} - 10.9\right)}_{\mathcal{K}} \right] \quad (2)$$

$$\frac{d\sigma}{dT} = -0.455 - 0.155 \left[\ln(1 + \mathcal{K}) - \frac{\mathcal{K}}{1 + \mathcal{K}} \cdot \frac{29300}{T} \right] \quad (3)$$

1.4 Explicit Calculation at $T = 1873 \text{ K}$

The adsorption parameter $\mathcal{K}$ and resulting surface tension are computed below for the two simulated cases.

Pure 316L ($\eta = 0$, $w_C = 0.030 \text{ wt}\%$, $w_O = 0.001 \text{ wt}\%$):

$$\begin{aligned} \mathcal{K} &= 10^{-0.17(0.001) - 0.427(0.030)} \times 0.001 \times e^{29300/1873 - 10.9} = 0.1114 \\ \sigma &= 1925 - 0.455(1873 - 1808) - 0.155(1873) \ln(1.1114) = 1864.8 \text{ mN m}^{-1} \\ \frac{d\sigma}{dT} &= -0.455 - 0.155 \left[\ln(1.1114) - \frac{0.1114}{1.1114} \cdot \frac{29300}{1873} \right] = -0.228 \text{ mN m}^{-1} \text{K}^{-1} \end{aligned}$$

316L + PLC ($\eta = 0.1$, $w_C = 0.120 \text{ wt}\%$, $w_O = 0.0411 \text{ wt}\%$):

$$\begin{aligned} \mathcal{K} &= 10^{-0.17(0.0411) - 0.427(0.120)} \times 0.0411 \times e^{29300/1873 - 10.9} = 4.127 \\ \sigma &= 1925 - 0.455(1873 - 1808) - 0.155(1873) \ln(5.127) = 1420.9 \text{ mN m}^{-1} \\ \frac{d\sigma}{dT} &= -0.455 - 0.155 \left[\ln(5.127) - \frac{4.127}{5.127} \cdot \frac{29300}{1873} \right] = +1.244 \text{ mN m}^{-1} \text{K}^{-1} \end{aligned}$$

1.5 Sensitivity to Dissolution Efficiency

Table 1: Surface tension at $T = 1873$ K (500 μm melt pool). Bold: adopted case.

η	w_C (wt%)	w_O (wt%)	σ (mN m^{-1})	$d\sigma/dT$ ($\text{mN m}^{-1}\text{K}^{-1}$)	Marangoni
0 (pure 316L)	0.030	0.0010	1864.8	-0.228	outward
0.001	0.031	0.0014	1853.3	-0.150	outward
0.01	0.039	0.0050	1767.5	+0.341	inward
0.02	0.048	0.0090	1696.3	+0.642	inward
0.05	0.075	0.0211	1555.1	+1.037	inward
0.1	0.120	0.0411	1420.9	+1.244	inward
1.0	0.921	0.397	1076.6	+1.388	inward

2 Simulation Cases

Two cases are run with laserbeamFoam on 32 MPI processes (hierarchical decomposition $2 \times 2 \times 8$ along $x \times y \times z$), differing only in surface tension properties.

Table 2: Simulation parameters. Surface tension properties differ between the two cases (bold rows); all other parameters are identical.

Parameter	Pure 316L	316L + PLC ($\eta = 0.1$)
Domain ($x \times y \times z$)	0.32 $\times$ 0.50 $\times$ 0.50 mm	
Base mesh	32 $\times$ 50 $\times$ 50 cells (10 μm)	
AMR max refinement	1 level $\rightarrow$ 5 μm minimum	
Max cells	200 000	
Laser power	200 W	
Laser radius	35 μm	
Laser scan speed	200 mm s^{-1}	
Laser path (z)	0.10 mm $\rightarrow$ 0.40 mm (0.30 mm track)	
Simulation time	1.5 ms	
Initial Δt	10^{-8} s	
Max Δt	2×10^{-6} s ($\text{Co} \leq 0.1$)	
Output interval	10^{-5} s (150 snapshots)	
MPI processes	32 (hierarchical $2 \times 2 \times 8$, order $y-x-z$)	
σ (N m^{-1})	1.8648	1.4209
$d\sigma/dT$ ($\text{N m}^{-1}\text{K}^{-1}$)	-2.28×10^{-4}	$+1.244 \times 10^{-3}$
Marangoni flow	outward	inward

References

- [1] K. Morohoshi et al., “Effects of Carbon and Oxygen on Fe–C–O Melt Surface Tension,” *ISIJ Int.*, vol. 53, no. 8, pp. 1315–1319, 2013.